Quantitative Automated Reasoning with Satisfiability Modulo Theories

Synopsis

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Automated reasoning has significantly enhanced the reliability and trustworthiness of modern computing systems. However, existing reasoning tools remain limited when dealing with systems that incorporate randomness or operate under uncertainty. Such settings naturally demand *quantitative* reasoning frameworks capable of providing principled guarantees that account for the underlying uncertainty. Satisfiability Modulo Theories (SMT) has emerged as one of the most powerful and widely used reasoning paradigms. In this thesis, we advance SMT beyond its traditional qualitative boundaries and develop new methodologies that enable *quantitative reasoning* within the SMT framework.

The SMT framework encompasses a rich landscape of theories, each supporting distinct variable types and constraint languages, to enable reasoning across a wide variety of systems. As we extend SMT toward quantitative analysis, the nature of the quantitative queries themselves depends closely on the underlying theory and the application domain. In this thesis, we study the key quantitative questions that surface across SMT theories, especially those that are directly motivated by application needs.

The interplay between SMT theories and real-world applications gives rise to several distinct forms of quantitative analysis. For example, quantitative software verification, automating cryptographic attacks, and bug localization require *counting solutions over bit-vector* domains. Applications such as network reliability under power-flow models and probabilistic model checking mix discrete and continuous variables and ask for counts projected onto discrete variables. Problems such as fairness and robustness quantification for AI

systems require volume computation over the continuous domain of linear real arithmetic. In the presence of function specifications, we are often interested in how many different functions can be implemented that satisfy the specifications. *In this thesis, we address these four classes of quantitative questions.*

We begin with the case when the theory is *discrete*, for example, the theory of quantifier-free bit-vectors. In 2014, Hadarean et al. [Had+14] observed that, in many cases, eager bit-blasting is the best approach for SMT bit-vector problems. Prior approaches to counting bit-vector formulas exploited structural properties of the formula. In [SM24a], we proposed a tool that leverages eager equi-cardinal bit-blasting, followed by simplification of the formula and leveraging Boolean model counters. With a careful combination of these three components, we achieve 4× performance improvement over the current state-of-the-art for bitvector model counting.

Moving beyond purely discrete settings, the second class of problems arises in *mixed discrete-continuous domains*, requiring counts projected onto discrete variables. In [SM25], we adapt hashing-based counting, initiated by Chakraborty, Meel, and Vardi [CMV13] and extended to SMT by Chistikov, Dimitrova, and Majumdar [CDM17], to this hybrid setting, yielding a theory-aware reduction that preserves scalability while focusing on the discrete projection. The key idea for scalability for this set of problems lies in using a correct hash function and employing an efficient solver for hash functions. It shows 5× performance improvement over the current state-of-the-art.

Our third setting deals with quantitative reasoning over *purely continuous domains*, specifically SMT formulas over linear real arithmetic. Here, the quantitative question becomes the *volume* of the feasible region. Prior work by Ge [Ge24] decomposes the region into *disjoint* polytopes, approximates each polytope's volume, and sums the results, but enforcing disjointness can cause a prohibitive explosion in the number of polytopes. In [SSM25], we address this bottleneck by decomposing into *non-disjoint* polytopes and aggregating via a principled union-of-sets formulation inspired by Chakraborty, Vinodchandran, and Meel [CVM22], reducing region blow-up while retaining accuracy guarantees. Our tool solves 8× more problems than the current state-of-the-art.

Finally, we address quantitative reasoning over the *space of functions*, a setting that pushes beyond variable assignments into functional domains. Functions play a significant part in SMT in terms of uninterpreted functions. Also, beyond satisfiability, SMT is also used for synthesis: given a specification, synthesize a function that satisfies it. The corresponding quantitative question asks how many satisfying functions exist. In [SJM24], we take a first step towards solving this question by counting Boolean specifications. The problem is particularly challenging, as synthesizing even one function is a hard problem. We design an algorithm, SkolemFC, that approximates the number of functions without synthesizing even one function. SkolemFC counts Skolem functions using only $O(n \log n)$ SAT calls, and shows a performance at the scale of synthesizing one function.

All our algorithms come with theoretical guarantees of (ε, δ)-style accuracy. Our algorithms are realized in four user-friendly tools: pact for hybrid discrete-projected counting, ttc for LRA volume, csb for bit-vector counting (including projected, weighted, and sampling variants), and skolemfc for Skolem-function counting. Across standard suites, these tools achieve 5×-10× improvements over prior state of the art. Across these projects, we deliberately employ different algorithmic strategies for closely related problems; a central takeaway, echoing the broader SMT solving experience, is that no single technique dominates all quantitative tasks.

Having developed these counting engines, we demonstrate their practical reach by applying them to *probabilistic model checking*. When a system exhibits probabilistic behavior, the model checking question becomes quantitative: what is the reachability probability in a Markov chain? Modern probabilistic model checkers perform extremely well on many benchmarks, yet as Holtzen et al. [Hol+21] identify, there are classes of surprisingly simple models on which they struggle due to state-space explosion. In [SJM26], we show that such cases admit a natural reduction to bit-vector model counting: the relevant execution paths are encoded as a bitvector formula, and the overall verification task is decomposed into a set of structurally simple model-counting subproblems. Using an extened version of csb [SM24a] as the counting backend, our probabilistic model checking tool MC3 handles up to 32× more parallel processes than existing probabilistic model checkers on a collection of challenging benchmarks.

The SMT tools developed in this thesis are a modest attempt to expand the frontier of quantitative reasoning. Each tool was born from a concrete real-world need, yet once built, reached further than the problem that inspired it - as the application to probabilistic model checking illustrates. This is, perhaps, a small glimpse of the *virtuous cycle* Cook described: automated reasoning in critical application areas drives investment in foundational tools, while improvements in those tools drive further applications [Coo22]. The four tools presented here are early steps toward a much larger edifice; as the systems we build grow more probabilistic, and more uncertain, we hope this cycle will only spin faster.

Chapterwise Content

The thesis is organized into three parts, structured to move from theoretical foundations, through algorithmic contributions for each quantitative domain, to a real-world application, before concluding with a synthesis of lessons learned.

Part I: Orientation and Unifying Framework. Chapter 1 introduces the technical background, motivates the problem, and characterizes its inherent hardness across SMT theories. Chapter 2 casts all four quantitative settings under a common measure-theoretic framework, revealing structural commonalities and divergences across domains, and surveys the algorithmic design choices that inform the rest of the thesis.

Part II: Algorithms for Different SMT Counting Problems. Each chapter addresses one subproblem of SMT counting and is self-contained, as the algorithmic techniques across domains differ substantially. The chapters progress naturally from discrete to hybrid to continuous to functional domains. Chapter 3 covers the discrete bit-vector setting. Chapter 4 addresses hybrid discrete-continuous domains. Chapter 5 handles volume computation over linear real arithmetic. Chapter 6 treats counting in the functional domain of Skolem functions. Each chapter presents the algorithm, theoretical (ε, δ)-style correctness guarantees, and an experimental evaluation.

Part III: In the Wild and Concluding. Chapter 7 applies the counting engines developed in Part II to probabilistic model checking, demonstrating their practical reach on real-world benchmarks. Chapter 8 concludes the thesis by synthesizing the algorithmic lessons across all domains, serving as a practical reference for future work in quantitative SMT.

Selected Publications

[SJM24] Arijit Shaw, Brendan Juba, and Kuldeep S. Meel. "An Approximate Skolem Function Counter". In: *Proceedings of AAAI Conference on Artificial Intelligence (AAAI)*. 2024.

[SJM26] Arijit Shaw, Sebastian Junges, and Kuldeep S. Meel. "Covering-based Approximate Model Counting for Statistical Model Checking". In: *Under Submission at CAV*. 2026.

[SM24a] Arijit Shaw and Kuldeep S Meel. "CSB: A Counting and Sampling Tool for Bitvectors". In: *Proc. of SMT Workshop at CAV*. 2024.

[SM24b] Arijit Shaw and Kuldeep S. Meel. "Model Counting in the Wild". In: *Proceedings of International Conference on Knowledge Representation and Reasoning (KR)*. 2024.

[SM25] Arijit Shaw and Kuldeep S Meel. "Approximate SMT Counting Beyond Discrete Domains". In: *Proceedings of Design Automation Conference (DAC)*. 2025.

[SSM25] Arijit Shaw, Uddalok Sarkar, and Kuldeep S Meel. "Efficient Volume Computation for SMT Formulas". In: *Proceedings of Knowledge Representation and Reasoning (KR)*. 2025.

[Yan+23] Jiong Yang, Arijit Shaw, Teodora Baluta, Mate Soos, and Kuldeep S Meel. "Explaining SAT Solving Using Causal Reasoning". In: *Proceedings of the Theory and Applications of Satisfiability Testing (SAT)*. 2023.

Additional References

[CDM17] Dmitry Chistikov, Rayna Dimitrova, and Rupak Majumdar. "Approximate counting in SMT and value estimation for probabilistic programs". In: *Acta Informatica* 8 (2017).

[CMV13] Supratik Chakraborty, Kuldeep S Meel, and Moshe Y Vardi. "A scalable approximate model counter". In: *Proceedings of International Conference on Constraint Programming (CP)*. 2013.

[Coo22] Byron Cook. *Automated reasoning's scientific frontiers*. [link]. Amazon Science blog post. 2022.

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[Ge24] Cunjing Ge. "sharpSMT: A Scalable Toolkit for Measuring Solution Spaces of SMT(LA) Formulas". In: *Frontiers of Computer Science (FCS)* (2024).

[Had+14] Liana Hadarean, Kshitij Bansal, Dejan Jovanović, Clark Barrett, and Cesare Tinelli. "A Tale of Two Solvers: Eager and Lazy Approaches to Bit-Vectors". In: *Proceedings of International Conference on Computer-Aided Verification (CAV)*. 2014.

[Hol+21] Steven Holtzen, Sebastian Junges, Marcell Vazquez-Chanlatte, Todd Millstein, Sanjit A Seshia, and Guy Van den Broeck. "Model checking finite-horizon Markov chains with probabilistic inference". In: *Proceedings of International Conference on Computer-Aided Verification (CAV)*. 2021.